Topic: Solving systems three ways

Question: Which solution is not being done with substitution? The steps are not explained, so we'll need to figure out what was done in each step.

Answer choices:

A

$$y = x - 3$$
$$y = 4x - 9$$
$$x - 3 = 4x - 9$$
$$3x = 6$$
$$x = 2$$

B

$$y = x - 3$$
$$y = 4x - 9$$
$$0 = -3x + 6$$
$$3x = 6$$
$$x = 2$$

C

$$y = x - 3$$
$$y = 4x - 9$$
$$x = y + 3$$
$$y = 4(y + 3) - 9$$
$$y = 4y + 12 - 9$$
$$y = 4y + 3$$
$$-3y = 3$$
$$y = -1$$

D

$$y = x - 3$$
$$y = 4x - 9$$
$$x = \frac{y + 9}{4}$$
$$y = \frac{y + 9}{4} - 3$$
$$4y = y + 9 - 12$$
$$3y = -3$$
$$y = -1$$

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Solution: B

Looking at answer choice B,

$$\begin{array}{r} y = x - 3 \\ y = 4x - 9 \\ \hline 0 = -3x + 6 \end{array}$$

this first step consisted of subtracting the second equation from the first equation, so this is solving by elimination, not substitution.

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Topic: Solving systems three ways

Question: To solve the system by elimination, which of these would not be a useful first step?

$$5x + y = 13$$

$$x - 2y = 7$$

Answer choices:

A Multiply the second equation by -5.

B Multiply the first equation by 2.

C Subtract the second equation from the first.

D Divide the second equation by 2.

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Solution: C

If we did the step in answer choice A, we'd have the following system of equations:

$$5x + y = 13$$

$$-5x + 10y = -35$$

We could then add the two equations and eliminate x. So this would be a useful first step.

If we did the step in answer choice B, we'd have the following system of equations:

$$10x + 2y = 26$$

$$x - 2y = 7$$

We could then add the two equations and eliminate y. So this would be a useful first step.

If we did the step in answer choice C, we'd have the following equation:

$$5x + y - x + 2y = 13 - 7$$

$$4x + 3y = 6$$

At this point there's no single step that could be done to eliminate x or y, so this wouldn't be a useful first step.

If we did the step in answer choice D, we'd have the following system of equations:

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$$5x + y = 13$$

$$\frac{1}{2}x - y = \frac{7}{2}$$

We could then add the two equations and eliminate y. So this would be a useful first step.

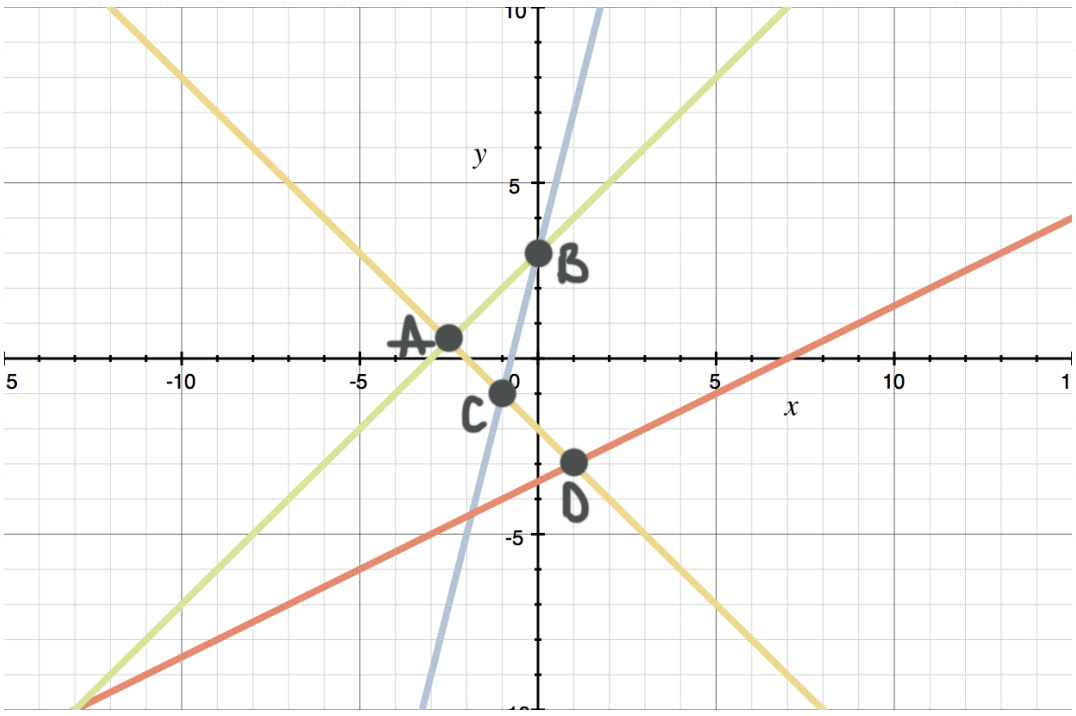
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Topic: Solving systems three ways

Question: Which point in the graph is the solution to the system?

$$4x - y = -3$$

$$x + y = -2$$



Answer choices:

A Point A

B Point B

C Point C

D Point D

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Solution: C

We can rewrite the two equations in slope-intercept form as

$$4x - y = -3$$

$$y = 4x + 3$$

and

$$x + y = -2$$

$$y = -x - 2$$

The equation $y = 4x + 3$ has a slope of 4 and a y-intercept of 3, which means it's represented by the blue line, and $y = -x - 2$ has a slope of -1 and a y-intercept of -2, which means it's represented by the yellow line.

The blue and yellow lines intersect at point C.

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